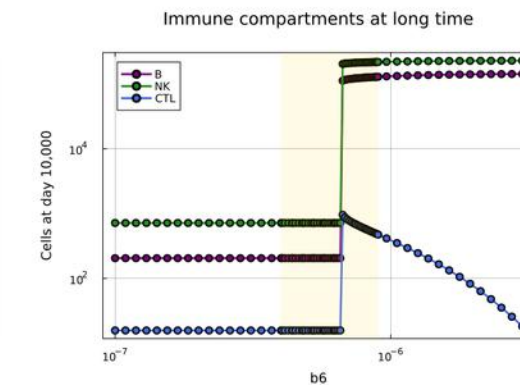
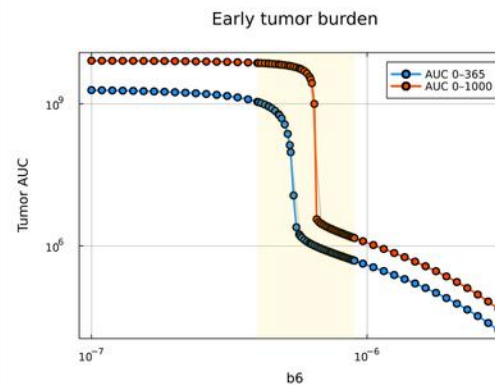
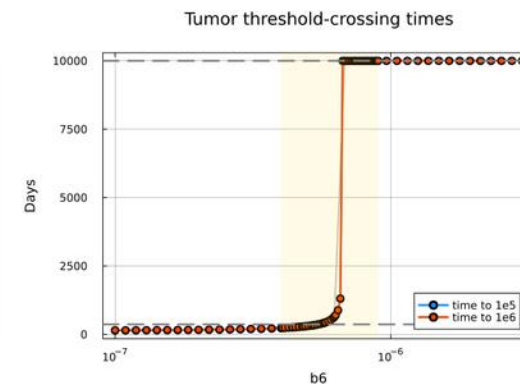
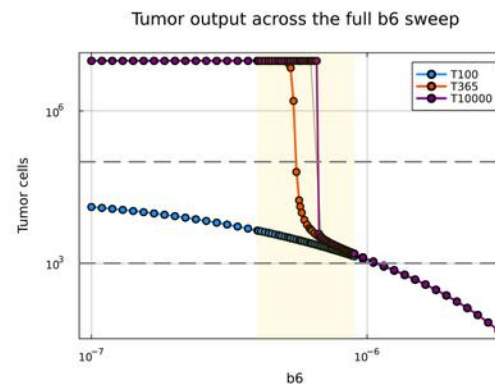


Long-Term ODE Dense b6 Sweep

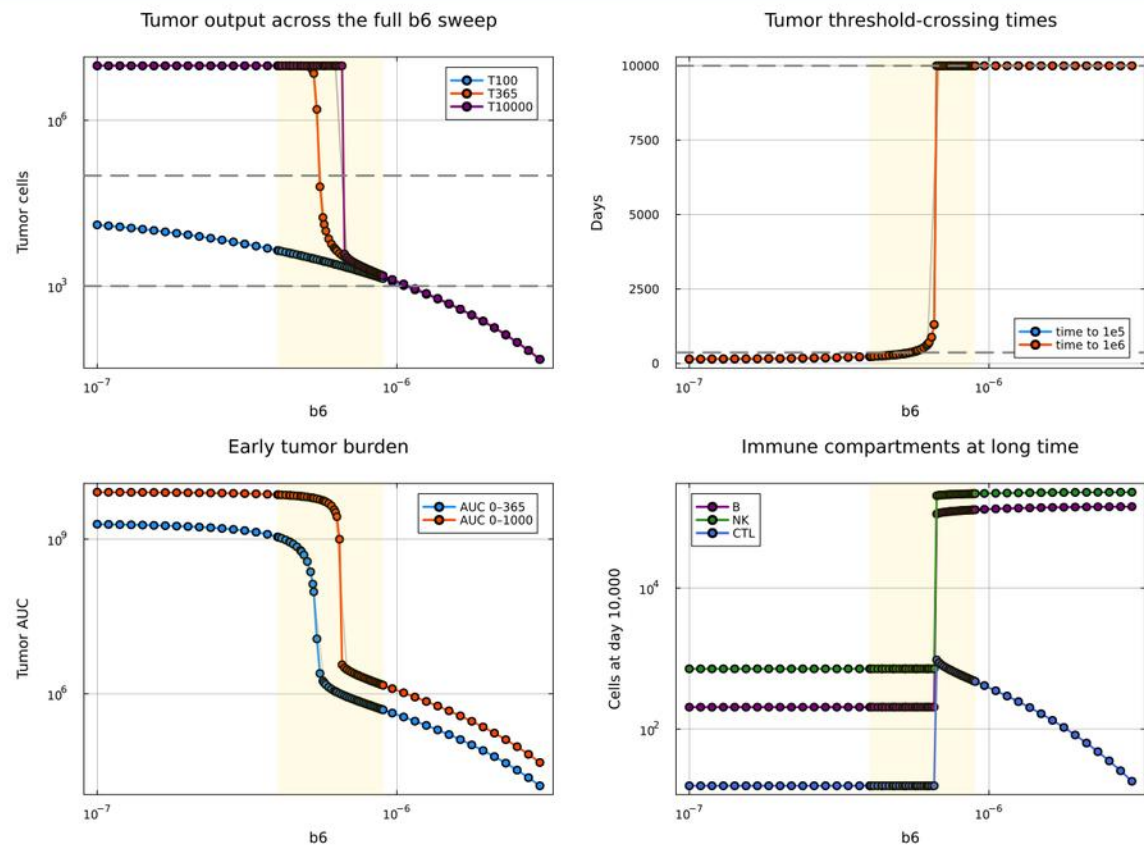
Model fixed after the meeting; only b6 is varied and each run is extended to 10,000 days

- Script in focus:
b6_dense_step0_step1_pretty.jl
- Baseline setup: direct CTL, b5 = 1e-4, baseline initial condition
- Main question: when does stronger B-cell killing create durable control instead of only delayed escape?



Finding 1: larger b6 strongly suppresses early tumor growth

Full 10,000-day sweep summary



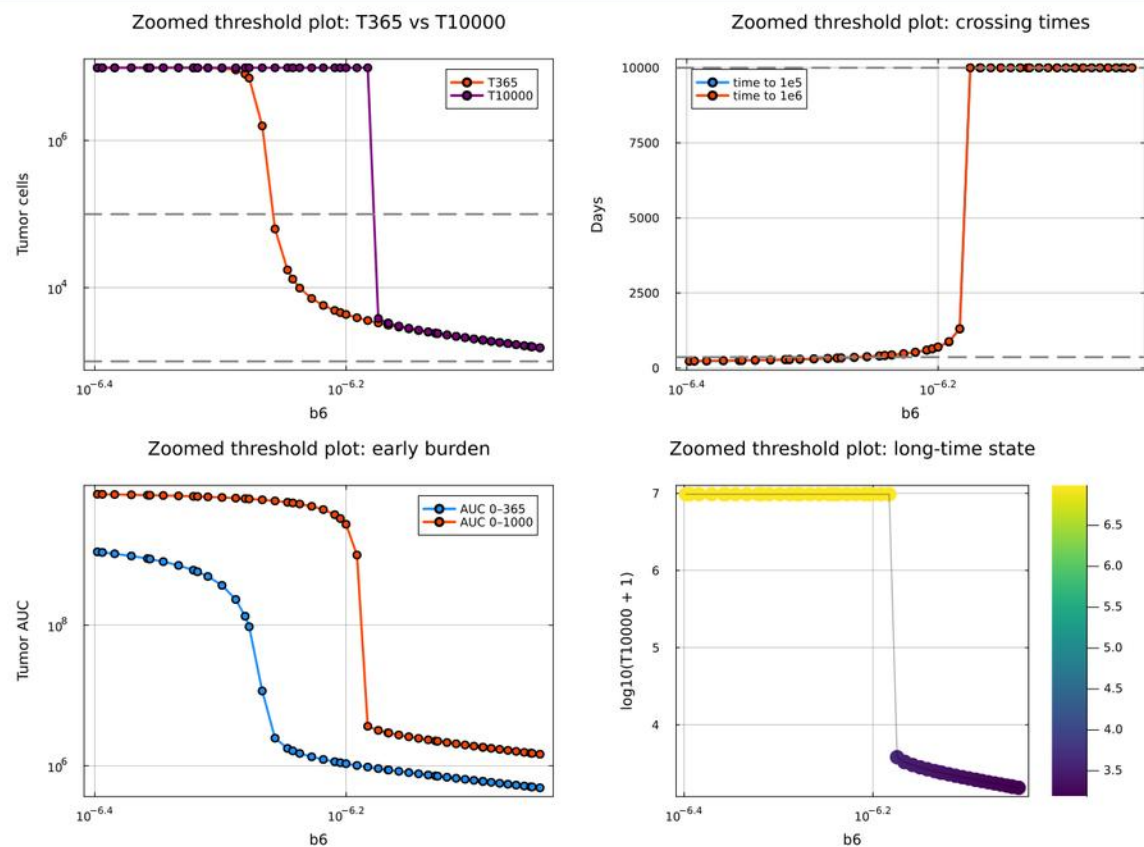
Main points

- T100 drops from 1.28×10^4 at $b_6 = 10^{-7}$ to 47 at $b_6 = 3 \times 10^{-6}$.
- Time to 10^5 shifts from 136 days to not reached by 10,000 days.
- AUC(0-365) falls from 1.95×10^9 to 1.63×10^4 , a drop of more than five orders of magnitude.

Early burden and threshold-crossing times both improve monotonically as b_6 increases.

Finding 2: long-time behavior changes in a narrow b_6 window

Threshold zoom around the switch region



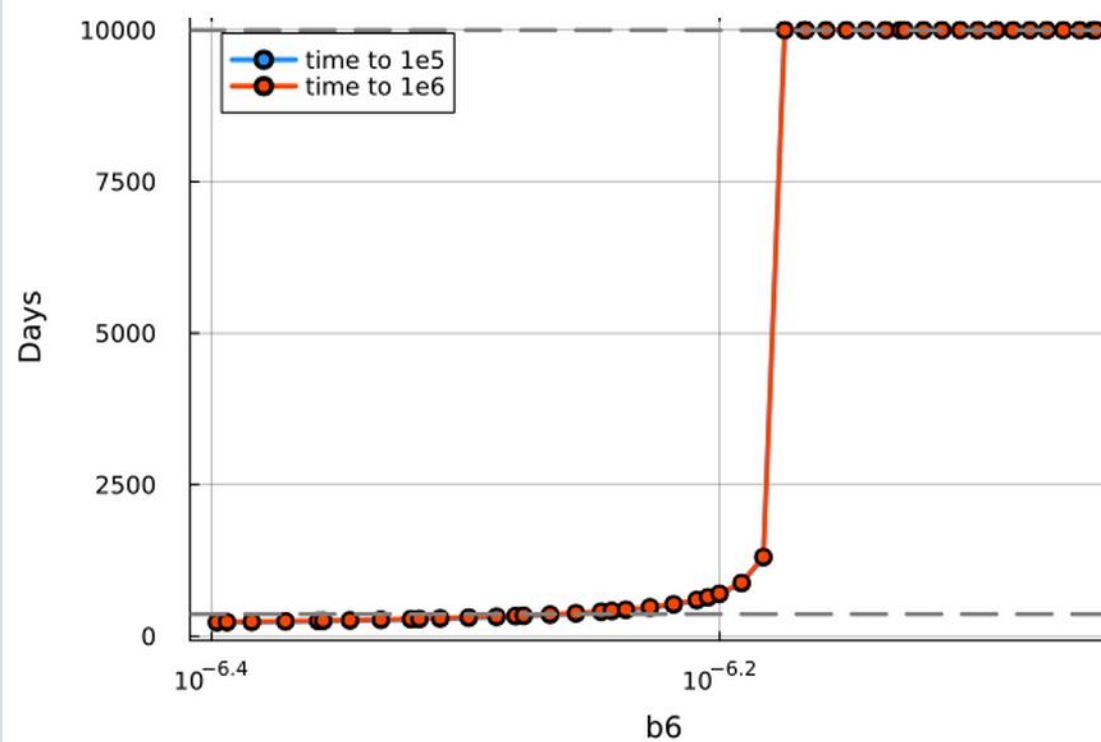
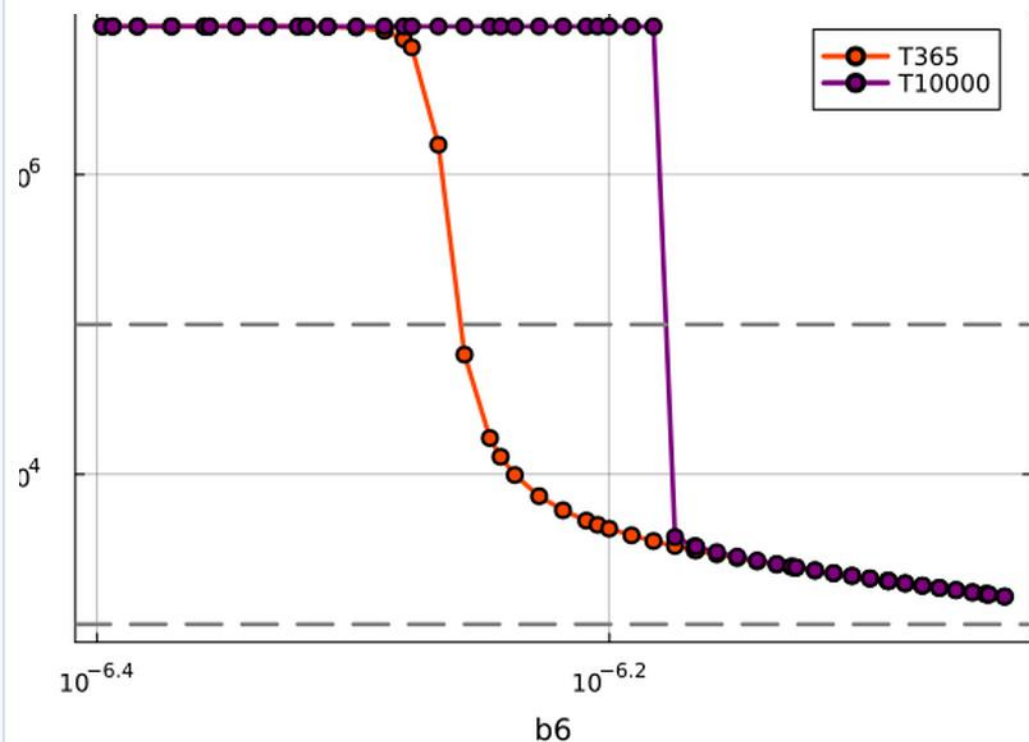
Main points

- Tumor-dominant solutions persist up to about $6.56e-7$.
- An intermediate or dormancy-like regime appears from about $6.69e-7$ to $1.05e-6$.
- Clear tumor-controlled solutions begin around $1.15e-6$ and continue through the top of the sweep.

The control boundary is not broad; it is concentrated near $b_6 \sim 10^{-6}$.

Finding 3: day 365 can look good even when long-time escape still occurs

Large-panel view of the threshold-region curves



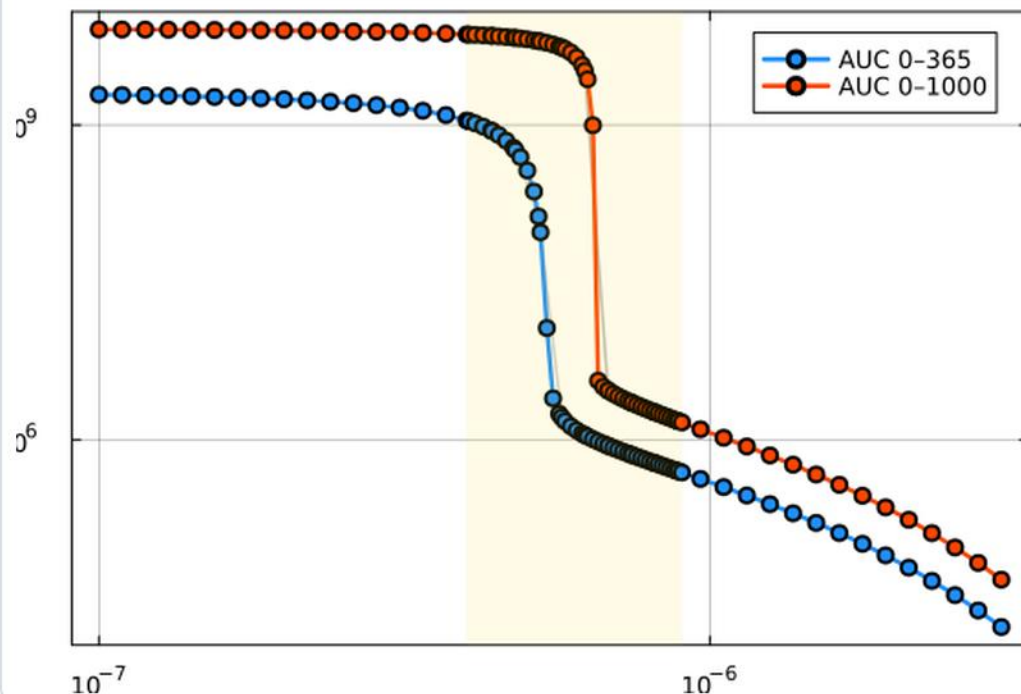
Takeaway

- At $b_6 \approx 6.05 \times 10^{-7}$, $T_{365} = 5.76 \times 10^3$ but T_{10000} rebounds to 9.74×10^6 .
- $T_{365} < 10^5$ starts near 5.54×10^{-7} ; $T_{10000} < 10^5$ starts later, near 6.69×10^{-7} .
- Long runs separate delay from durable control.

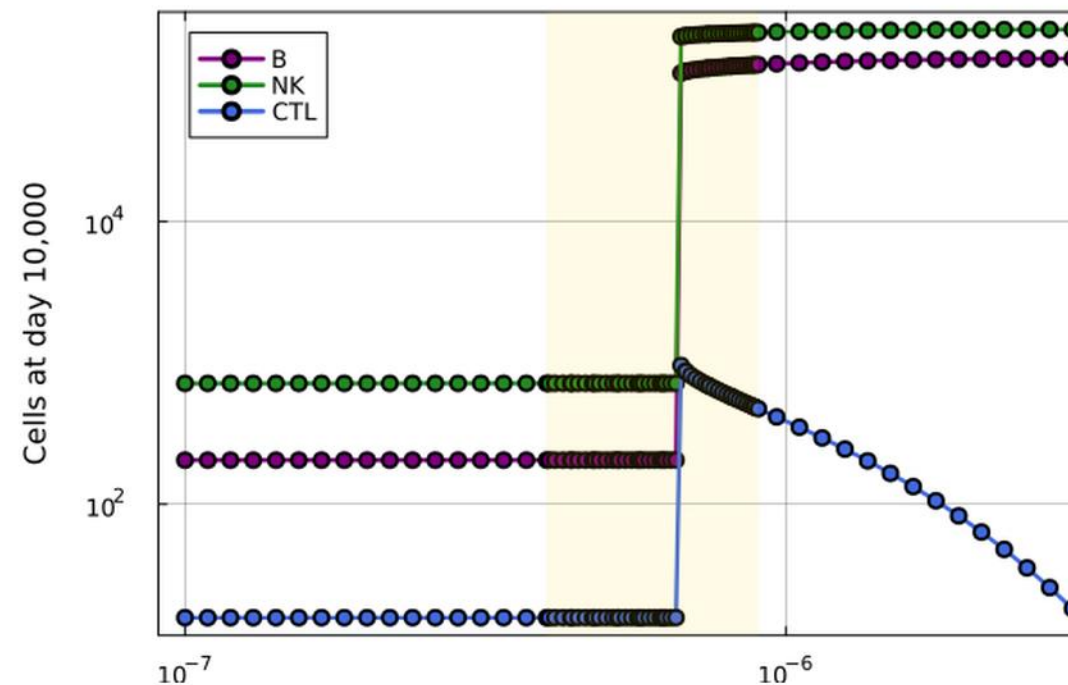
Finding 4: crossing the threshold reshapes the immune steady state

Early-burden collapse and long-time immune-state expansion

Early tumor burden



Immune compartments at long time



Takeaway

- B at day 10,000 rises from about 205 at $b_6 = 1e-7$ to $1.42e5$ at $b_6 = 3e-6$.
- NK at day 10,000 rises from about 716 to $2.28e5$ across the same range.
- CTL peaks near $b_6 = 6.69e-7$ at about 960, then falls once tumor becomes very small.

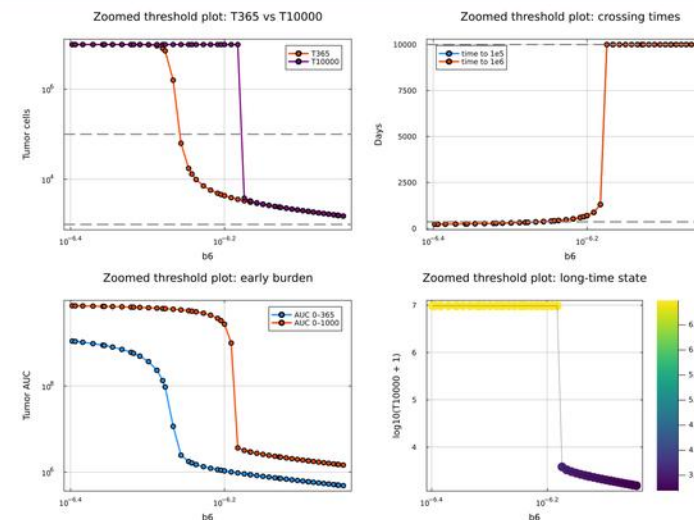
The long-time control state is associated with a high-B and high-NK background, not simply with maximal CTL.

Questions to ask the leading PhD

Suggested discussion prompts for the meeting

Quick numerical anchors

- Which endpoint should define control for this project: T365, T1000, T10000, or a threshold-crossing rule?
- Should the intermediate regime be interpreted as biologically meaningful dormancy, or do we treat it as a model artifact until validated?
- Is the estimated control window around $b_6 \sim 10^{-6}$ biologically plausible for the intended B-cell mechanism?
- Do we want the next long-time scan to focus on $b_5 \times b_6$, since B and NK change sharply across the threshold?



First T10000 < 1e5: $b_6 \approx 6.69e-7$

First T10000 < 1e3: $b_6 \approx 1.15e-6$

Transition window: $\sim 6.7e-7$ to $\sim 1.15e-6$

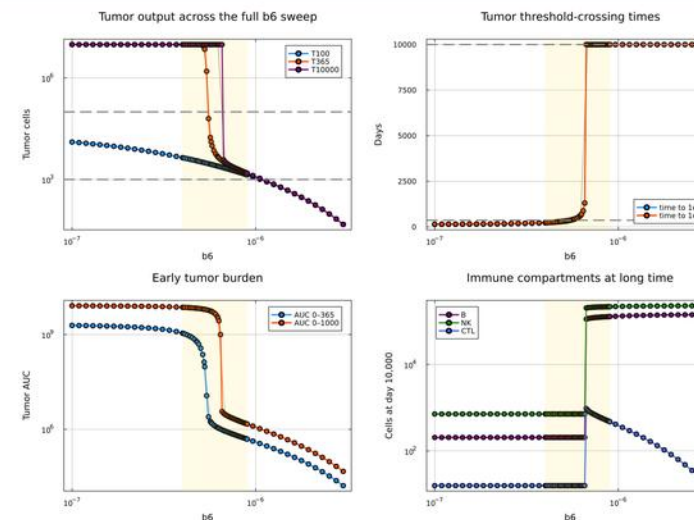
CTL peak: ~ 960 near $b_6 = 6.69e-7$

Future directions

Reasonable next steps after the long-term ODE result

Short recommendation

- Run a long-time $b_5 \times b_6$ sweep to see how the control boundary moves when MDSC suppression of B cells changes.
- Add focused SDE only around $6.5e-7$ to $1.2e-6$ to test whether the ODE threshold survives stochasticity.
- Check sensitivity to initial tumor seed and to the definition of control versus delay.
- If the threshold remains sharp, use continuation or bifurcation analysis to explain the switch mechanistically.



Best immediate next step: long-time $b_5 \times b_6$

Best robustness check: threshold-window SDE

Best interpretation check: redefine control criteria

Best mechanism study: bifurcation near $b_6 \sim 10^{-6}$